Digital Image Processing Lab (NCSH 302) **Session: Monsoon 2026-27**
B. Tech. 5th Sem (CSE-Hons) **Date: 01.09.2026**

Assignment-4 (Image Enhancement-III)

1. A factory uses an 8-bit grayscale camera to inspect polished metal sheets. Scratches and cracks appear as dark pixels. For the given grayscale image $f(x, y)$, calculate the defect threshold as, $T = I_{min} + \frac{1}{3} \times (I_{max} - I_{min})$ where I_{min} and $I_{\max}$ are the minimum and maximum pixel intensities. Pixels with intensities from I_{min} to T are considered suspected defect pixels. Use zero-based (x, y) coordinates, where x is the row and y is the coloumn.

   Generate the binary defect mask $B(x, y)$ using the condition below.

$$B(x,y) = 1, \quad if\ I_{min} \leq f(x,y) \leq T;$$
$$= 0, \quad otherwise.$$

   Also generate an enhanced image in which every detected defect pixel is assigned 255, while every background pixel retains its original intensity.

   Next, slide a 3×3 all-one mask over the complete binary defect mask $B(x, y)$ with stride 1 and no padding. At each valid top-left position (i, j), calculate

$$S(i,j) = sum\ (u = 0\ to\ 2\ and\ v = 0\ to\ 2\ of\ B\ (i + u, j + v))$$

   Declare the 3×3 block completely defective only when $S\ (i,j) = 9$. Because no padding is used, an $H \times W$ binary mask produces an $(H-2) \times (W-2)$ block-sum output. Count overlapping defective blocks using $N_{blocks} = sum\ (D\ (i,j))$, where

$$D\ (i,j) = 1,\ if\ S\ (i,j) = 9;$$
$$= 0,\ otherwise.$$

[15 Marks]

Sample Input:

8×8 Grayscale Image

30	35	40	45	50	100	110	120
25	32	38	44	52	105	115	125
20	28	34	42	48	110	120	130
22	29	36	43	55	115	125	135
100	110	120	130	140	120	130	140
105	115	125	135	145	125	135	145
110	120	130	140	150	130	30	40
115	125	135	145	150	135	45	55

Sample Output:

Threshold $T = 63.33$

Binary defect mask $B(x, y)$

1	**1**	**1**	**1**	**1**	0	0	0
1	**1**	**1**	**1**	**1**	0	0	0
1	**1**	**1**	**1**	**1**	0	0	0
1	**1**	**1**	**1**	**1**	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	**1**	**1**
0	0	0	0	0	0	**1**	**1**

Enhanced image (defect $pixels = 255$; background preserved)

255	255	255	255	255	100	110	120
255	255	255	255	255	105	115	125
255	255	255	255	255	110	120	130
255	255	255	255	255	115	125	135
100	110	120	130	140	120	130	140
105	115	125	135	145	125	135	145
110	120	130	140	150	130	255	255
115	125	135	145	150	135	255	255

Valid 3×3 block-sum output $S(i, j), size\ 6 \times 6$

9	**9**	**9**	6	3	0
9	**9**	**9**	6	3	0
6	6	6	4	2	0
3	3	3	2	1	0
0	0	0	0	1	2
0	0	0	0	2	4

Completely defective-block map $D(i,j), size\ 6 \times 6$

1	**1**	**1**	0	0	0
1	**1**	**1**	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

The top-left 4 × 5 defect region contains (4-3+1) × (5-3+1) = 2 × 3 = 6 overlapping 3 × 3 defective blocks.

Thus, $N_{blocks} = 6$.

2. An autonomous vehicle captures a low-contrast 8-bit grayscale road image at night in which lane markings and obstacle boundaries appear blurred, while sensor noise and bright reflections may produce false edges. The input image is provided as "night_road.jpg". Read the image, resize the image to 512 × 512 pixels, convert it to grayscale and perform the following tasks.

(a) Edge Sharpening using Sobel operator:

Apply the first-order Sobel operators

$$G_x = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad and \quad G_y = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Calculate the gradient magnitude using the Sobel operator as $G = |G_x| + |G_y|$. Determine the 90th percentile P_{90} of gradient magnitude values and use this value as threshold to generate the Sobel-sharpened image. Implement all convolution operations manually using zero-padding.

[10 Marks]

(b) Edge Sharpening using Laplacian operators:

Apply the four-neighbor and eight-neighbor Laplacian operators

$$L_4 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad and \quad L_8 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Generate the corresponding sharpened images using $gL_4 = f - L_4(f)\ and\ gL_8 = f - L_8(f)$. Determine the 90th percentile P_{90} of gradient magnitude values and use this value as threshold to generate the Laplacian-sharpened images. Implement all convolution operations manually using zero-padding, clip the resulting values to $[0, 255]$, and convert the outputs to 8-bit images.

[10 Marks]

(c) **Comparison of Sharpening Methods:**

For Sobel-sharpened image, four-neighbour Laplacian-sharpened image, and eight-neighbour Laplacian-sharpened image, calculate the Mean Gradient Magnitude as:

$$M_G = \frac{1}{HW} \sum_{x=0}^{H-1} \sum_{y=0}^{W-1} |G_g(x, y)|$$

where H and and $W$are the height and weight of the sharpened image and $|G_g(x, y)|$ is the absolute gradient value at pixel location (x, y) of the sharpened image. A higher M_G indicates that the image contains stronger edges.

Compare the M_G values of all three sharpened images and identify the sharpening method that produces the highest M_G value.

[5 Marks]

Sample Input:

Resized 512 x 512 grayscale Image



Sample Output:

a) P10 = 0.0000
   P90 = 80.0000



b)



c)

Image	**Mean Gradient Magnitude (M_G)**
Sobel Sharpened	188.86
Four-Neighbour Laplacian Sharpened	23.06
Eight-Neighbour Laplacian Sharpened	82.53

Method with the highest M_G: Sobel